

Learning About the Long Run

L. Farmer, E. Nakamura, J. Steinsson

Giacomo Cattelan

BGGMP

March 22, 2022

Table of Contents

Introduction

Testing the RE Hypothesis

Model

Motivation

Anomalies of professional forecasts inconsistent with the Rational Expectation Hypothesis:

1. Biased;
2. Forecast errors are autocorrelated;
3. Forecast revisions predict future forecast errors.

Motivation

Anomalies of professional forecasts inconsistent with the Rational Expectation Hypothesis:

1. Biased;
2. Forecast errors are autocorrelated;
3. Forecast revisions predict future forecast errors.

Possible explanations:

- ▶ models of bounded rationality (e.g. Bordalo et al. 2018-2020);
- ▶ rationality BUT imperfect information acquisition (Sims 2003, Woodford 2003).

Motivation

Anomalies of professional forecasts inconsistent with the Rational Expectation Hypothesis:

1. Biased;
2. Forecast errors are autocorrelated;
3. Forecast revisions predict future forecast errors.

Possible explanations:

- ▶ models of bounded rationality (e.g. Bordalo et al. 2018-2020);
- ▶ rationality BUT imperfect information acquisition (Sims 2003, Woodford 2003).

Good for modeling hh and firm behavior, less realistic for professional forecasters and traders.

Motivation

More realistically, forecasters don't know the "true" model generating the data and they have to learn it via Bayes rule.

Motivation

More realistically, forecasters don't know the "true" model generating the data and they have to learn it via Bayes rule.

In a rich model with stationary and non-stationary components:

- ▶ LR dynamics may be independent of SR dynamics;

Motivation

More realistically, forecasters don't know the "true" model generating the data and they have to learn it via Bayes rule.

In a rich model with stationary and non-stationary components:

- ▶ LR dynamics may be independent of SR dynamics;
- ▶ SR dynamics is uninformative of LR trends;

Motivation

More realistically, forecasters don't know the "true" model generating the data and they have to learn it via Bayes rule.

In a rich model with stationary and non-stationary components:

- ▶ LR dynamics may be independent of SR dynamics;
- ▶ SR dynamics is uninformative of LR trends;
- ▶ evidence about the LR accumulates slowly and learning occurs slowly.

Motivation

More realistically, forecasters don't know the "true" model generating the data and they have to learn it via Bayes rule.

In a rich model with stationary and non-stationary components:

- ▶ LR dynamics may be independent of SR dynamics;
- ▶ SR dynamics is uninformative of LR trends;
- ▶ evidence about the LR accumulates slowly and learning occurs slowly.

Bayesian learning can explain persistence of forecasts anomalies.

Table of Contents

Introduction

Testing the RE Hypothesis

Model

Data

- ▶ Variable of interest: 3-months T-bill rate.
Source: 3-months T-bill rate on the secondary market from the Board of Governors, sample 1951:Q2-2019:Q4.
Forecasts source: SPF from the Philadelphia FED, sample 1981:Q3-2019:Q4.

Data

- ▶ Variable of interest: 3-months T-bill rate.
Source: 3-months T-bill rate on the secondary market from the Board of Governors, sample 1951:Q2-2019:Q4.
Forecasts source: SPF from the Philadelphia FED, sample 1981:Q3-2019:Q4.
- ▶ Variable of interest: real GDP growth.
Source: Philadelphia FED Real-Time DS; sample 1965-2019.
Forecasts source: Congressional Budget Office annual economic outlook with forecasts from 6 to 11 years; sample 1976-2019.

Assumption of the RE Hypothesis

1. Agents minimize MSE:

$$\mathbb{F}_t y_{t+h} = \mathbb{E}_t y_{t+h}$$

2. Agents know the true model of the world.

Assumption of the RE Hypothesis

1. Agents minimize MSE:

$$\mathbb{F}_t y_{t+h} = \mathbb{E}_t y_{t+h}$$

2. Agents know the true model of the world.

Let $e_{t+h|t} = y_{t+h} - \mathbb{F}_t y_{t+h}$.

$$H_0 : \mathbb{E}_t e_{t+h|t} = 0$$

Forecasting Anomalies

1. Bias:

$$e_{t+h|t} = \alpha + u_{t+h} \quad H_0 : \alpha = 0$$

Forecasting Anomalies

1. Bias:

$$e_{t+h|t} = \alpha + u_{t+h} \quad H_0 : \alpha = 0$$

2. Autocorrelation:

$$e_{t+h|t} = \alpha + \beta e_{t|t-h} + u_{t+h} \quad H_0 : \alpha = 0, \beta = 0$$

Forecasting Anomalies

1. Bias:

$$e_{t+h|t} = \alpha + u_{t+h} \quad H_0 : \alpha = 0$$

2. Autocorrelation:

$$e_{t+h|t} = \alpha + \beta e_{t|t-h} + u_{t+h} \quad H_0 : \alpha = 0, \beta = 0$$

3. Mincer-Zarnowitz Regression:

$$y_{t+h} = \alpha + \beta \mathbb{F}_t y_{t+h} + u_{t+h} \quad H_0 : \alpha = 0, \beta = 1$$

Forecasting Anomalies

1. Bias:

$$e_{t+h|t} = \alpha + u_{t+h} \quad H_0 : \alpha = 0$$

2. Autocorrelation:

$$e_{t+h|t} = \alpha + \beta e_{t|t-h} + u_{t+h} \quad H_0 : \alpha = 0, \beta = 0$$

3. Mincer-Zarnowitz Regression:

$$y_{t+h} = \alpha + \beta \mathbb{F}_t y_{t+h} + u_{t+h} \quad H_0 : \alpha = 0, \beta = 1$$

4. Coibion-Gorodnichenko Test:

$$e_{t+h|t} = \alpha + \beta (\mathbb{F}_t y_{t+h} - \mathbb{F}_{t-1} y_{t+h}) + u_{t+h} \quad H_0 : \alpha = 0, \beta = 0$$

Forecasting Anomalies

	Forecast Horizon				
	1	2	3	4	5
<i><u>Panel A: Bias</u></i>					
T-Bill	-0.18*** (0.06)	-0.34*** (0.11)	-0.52*** (0.16)	-0.70*** (0.20)	—
GDP Growth	0.27 (0.21)	-0.27 (0.34)	-0.54 (0.44)	-0.62 (0.47)	-0.52 (0.44)
<i><u>Panel B: Autocorrelation</u></i>					
T-Bill	0.30* (0.16)	0.27** (0.11)	0.24** (0.11)	0.13 (0.12)	—
GDP Growth	0.22 (0.16)	0.16 (0.17)	0.11 (0.14)	0.08 (0.18)	0.08 (0.11)
<i><u>Panel C: Mincer-Zarnowitz</u></i>					
T-Bill	0.97 (0.02)	0.94* (0.04)	0.90** (0.05)	0.86** (0.06)	—
GDP Growth	0.94 (0.12)	0.60 (0.31)	0.03*** (0.31)	-0.42*** (0.22)	-0.43*** (0.36)
<i><u>Panel D: Coibion-Gorodnichenko</u></i>					
T-Bill	0.23* (0.13)	0.34** (0.15)	0.62*** (0.18)	—	—
GDP Growth	0.08 (0.11)	0.00 (0.23)	0.50 (0.52)	-1.63*** (0.41)	-1.46*** (0.45)

Yield Curve Implications of RE

- Under RE

$$y_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} y_{t+j}^{(1)} + \text{risk premium}$$

Yield Curve Implications of RE

- Under RE

$$y_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} y_{t+j}^{(1)} + \text{risk premium}$$

- Campbell-Shiller regression:

$$\frac{1}{n} \sum_{j=0}^{n-1} y_{t+j}^{(1)} - y_t^{(1)} = \alpha + \beta(y_t^{(n)} - y_t^{(1)}) + u_t \quad H_0 : \beta = 1$$

Yield Curve Implications of RE

- Under RE

$$y_t^{(n)} = \frac{1}{n} \sum_{j=0}^{n-1} y_{t+j}^{(1)} + \text{risk premium}$$

- Campbell-Shiller regression:

$$\frac{1}{n} \sum_{j=0}^{n-1} y_{t+j}^{(1)} - y_t^{(1)} = \alpha + \beta(y_t^{(n)} - y_t^{(1)}) + u_t \quad H_0 : \beta = 1$$

- yield spread predicts increase in long run yield

$$y_{t+1}^{(n-1)} - y_t^{(n)} = \alpha + \beta \frac{1}{n-1} (y_t^{(n)} - y_t^{(1)}) + u_t \quad H_0 : \beta = 1$$

Yield Curve Implications of RE

	Long Horizon n						
	2	3	4	8	12	20	40
Future Short Rates	-0.01*** (0.22)	0.11*** (0.22)	0.18*** (0.21)	0.39** (0.25)	0.57 (0.27)	0.74 (0.23)	0.71 (0.22)
Change in Long Rate	-1.02*** (0.43)	-0.91*** (0.54)	-1.03*** (0.57)	-1.29*** (0.61)	-1.61*** (0.65)	-2.04*** (0.67)	-2.75*** (0.86)

Table of Contents

Introduction

Testing the RE Hypothesis

Model

Unobserved Components Model for the Nominal Short Rate

$$\begin{aligned}y_t &= \mu_t + x_t \\ \mu_t &= \mu_{t-1} + \sqrt{\gamma}\sigma\eta_t & \eta_t &\sim N(0, 1) \\ x_t &= \rho x_{t-1} + \sqrt{1-\gamma}\sigma\omega_t & \omega_t &\sim N(0, 1)\end{aligned}$$

Unobserved Components Model for the Nominal Short Rate

$$\begin{aligned}y_t &= \mu_t + x_t \\ \mu_t &= \mu_{t-1} + \sqrt{\gamma}\sigma\eta_t & \eta_t &\sim N(0,1) \\ x_t &= \rho x_{t-1} + \sqrt{1-\gamma}\sigma\omega_t & \omega_t &\sim N(0,1)\end{aligned}$$

$$\mathbb{F}_t y_{t+h} = \mu_t + \rho^h x_t \xrightarrow{h \rightarrow \infty} \mu_t$$

Unobserved Components Model for the Nominal Short Rate

$$\begin{aligned}y_t &= \mu_t + x_t \\ \mu_t &= \mu_{t-1} + \sqrt{\gamma}\sigma\eta_t & \eta_t &\sim N(0,1) \\ x_t &= \rho x_{t-1} + \sqrt{1-\gamma}\sigma\omega_t & \omega_t &\sim N(0,1)\end{aligned}$$

$$\mathbb{E}_t y_{t+h} = \mu_t + \rho^h x_t \rightarrow_{h \rightarrow \infty} \mu_t$$

$$y_t^{(n)} = c^{(n)} + \frac{1}{n} \sum_{h=0}^{n-1} \mathbb{E}_t y_{t+h} = c^{(n)} + \mu_t + \frac{1}{n} \sum_{h=0}^{n-1} \rho^h x_t$$

Learning

- ▶ Observable: y_t ; states/non-observable (μ_t, x_t) ; not known (ρ, γ, σ) ;

Learning

- ▶ Observable: y_t ; states/non-observable (μ_t, x_t) ; not known (ρ, γ, σ) ;
- ▶ learning starts 1952:Q2
 - ▶ different previous monetary regimes (gold standard and WWII effort);

Learning

- ▶ Observable: y_t ; states/non-observable (μ_t, x_t) ; not known (ρ, γ, σ) ;
- ▶ learning starts 1952:Q2
 - ▶ different previous monetary regimes (gold standard and WWII effort);
- ▶ as new data arrive:
 - ▶ update distribution of (ρ, γ, σ) with Gibbs Sampling Algorithm;
 - ▶ update path of (μ_t, x_t) for each value in parameter sample;
 - ▶ compute distribution of forecasts induced by each value in the parameter sample.

▶ Algorithm

Priors

Learning about (ρ, γ, σ^2) is "slow" $\implies$ prior is very persistent in the learning process.

Priors

Learning about (ρ, γ, σ^2) is "slow" $\implies$ prior is very persistent in the learning process.

$$\rho \sim N(\mu_\rho, \sigma_\rho) \quad \gamma \sim \text{Beta}(\alpha_\gamma, \beta_\gamma) \quad \sigma^2 \sim \text{IG}(\alpha_{\sigma^2}, \beta_{\sigma^2})$$

Priors

Learning about (ρ, γ, σ^2) is "slow" $\implies$ prior is very persistent in the learning process.

$$\rho \sim N(\mu_\rho, \sigma_\rho) \quad \gamma \sim \text{Beta}(\alpha_\gamma, \beta_\gamma) \quad \sigma^2 \sim \text{IG}(\alpha_{\sigma^2}, \beta_{\sigma^2})$$

Prior hyperparameters are searched to best generate the forecasting anomalies.

Priors

Learning about (ρ, γ, σ^2) is "slow" $\implies$ prior is very persistent in the learning process.

$$\rho \sim N(\mu_\rho, \sigma_\rho) \quad \gamma \sim \text{Beta}(\alpha_\gamma, \beta_\gamma) \quad \sigma^2 \sim \text{IG}(\alpha_{\sigma^2}, \beta_{\sigma^2})$$

Prior hyperparameters are searched to best generate the forecasting anomalies.

Learning about (μ, x) is "fast" $\implies$ prior fades away.

Priors

Learning about (ρ, γ, σ^2) is "slow" $\implies$ prior is very persistent in the learning process.

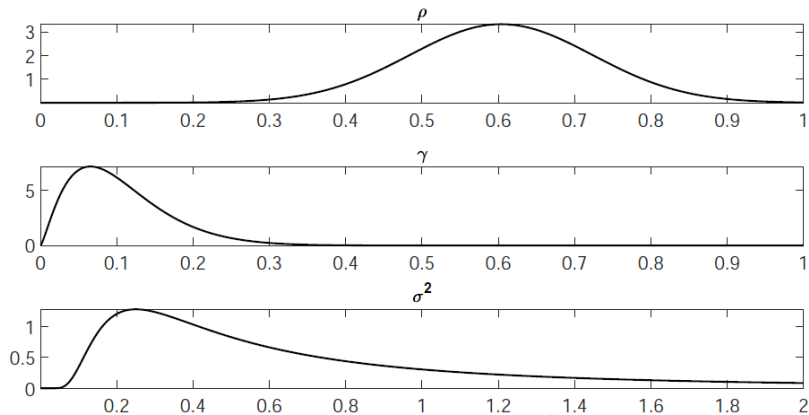
$$\rho \sim N(\mu_\rho, \sigma_\rho) \quad \gamma \sim \text{Beta}(\alpha_\gamma, \beta_\gamma) \quad \sigma^2 \sim \text{IG}(\alpha_{\sigma^2}, \beta_{\sigma^2})$$

Prior hyperparameters are searched to best generate the forecasting anomalies.

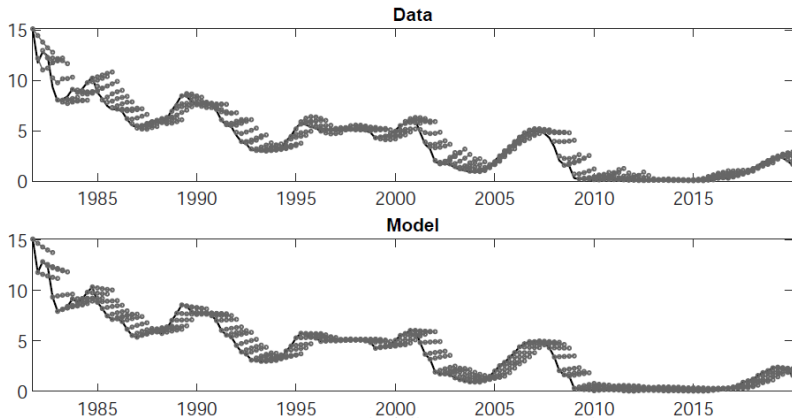
Learning about (μ, x) is "fast" $\implies$ prior fades away.

$$\mu_0 \sim N(y_{1952:Q2}, 1) \quad x \sim N(0, 1)$$

Marginal priors for (ρ, γ, σ^2)



Model Performance



► More

Model Performance

	Forecast Horizon			
	1	2	3	4
<i><u>Panel A: Bias</u></i>				
SPF	-0.18*** (0.06)	-0.34*** (0.11)	-0.52*** (0.16)	-0.70*** (0.20)
UC Model	-0.15** (0.06)	-0.27** (0.12)	-0.39** (0.17)	-0.50** (0.21)
<i><u>Panel B: Autocorrelation</u></i>				
SPF	0.30* (0.15)	0.27** (0.11)	0.24** (0.11)	0.13 (0.12)
UC Model	0.36** (0.16)	0.39*** (0.12)	0.35*** (0.11)	0.23** (0.11)
<i><u>Panel C: Mincer-Zarnowitz</u></i>				
SPF	0.97 (0.02)	0.94* (0.04)	0.90** (0.05)	0.86** (0.06)
UC Model	0.96* (0.02)	0.93** (0.03)	0.88** (0.05)	0.83*** (0.06)
<i><u>Panel D: Coibion-Gorodnichenko</u></i>				
SPF	0.23* (0.13)	0.34** (0.15)	0.62*** (0.18)	—
UC Model	0.38** (0.17)	0.55* (0.32)	0.88*** (0.34)	—

Model Performance

	Long Horizon n						
	2	3	4	8	12	20	40
<i>Panel A: Future Short Rates</i>							
Data	-0.01*** (0.22)	0.11*** (0.22)	0.18*** (0.21)	0.39** (0.25)	0.57 (0.27)	0.74 (0.23)	0.71 (0.22)
UC Model	-0.19*** (0.43)	0.01** (0.45)	0.14* (0.46)	0.58 (0.51)	0.83 (0.47)	0.94 (0.36)	0.95 (0.34)
<i>Panel B: Change in Long Rate</i>							
Data	-1.02*** (0.43)	-0.91*** (0.54)	-1.03*** (0.57)	-1.29*** (0.61)	-1.61*** (0.65)	-2.04*** (0.67)	-2.75*** (0.86)
UC Model	-1.08*** (0.72)	-1.11*** (0.73)	-1.14*** (0.74)	-1.28*** (0.80)	-1.42*** (0.87)	-1.72*** (1.00)	-2.36** (1.56)

Main Intuition

What happens if a "true" model is postulated?

Main Intuition

What happens if a "true" model is postulated?

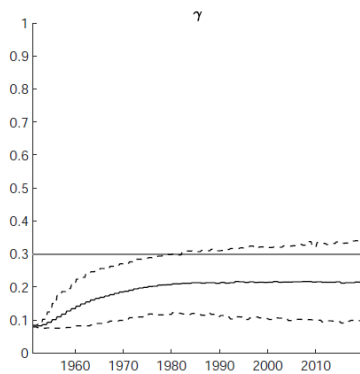
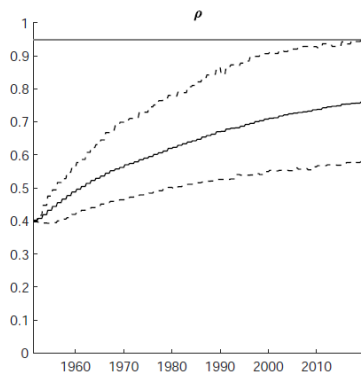
- ▶ Simulation of data from the UC model with true parameters roughly equal to the mean of beliefs in the second half of the sample: $\rho = 0.95$ $\gamma = 0.3$ and $\sigma = 0.5$;

Main Intuition

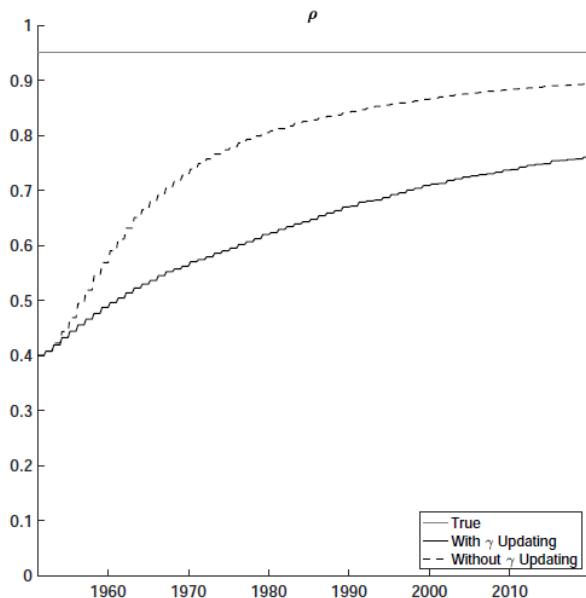
What happens if a "true" model is postulated?

- ▶ Simulation of data from the UC model with true parameters roughly equal to the mean of beliefs in the second half of the sample: $\rho = 0.95$ $\gamma = 0.3$ and $\sigma = 0.5$;
- ▶ perform Bayesian learning in the case of unbiased, downward biased and upward biased priors;
- ▶ only downward biased prior generates the anomalies in the data.

Dynamics of ρ and γ



Learning of ρ in the presence and absence of γ



To wrap up

New interpretation of forecasts anomalies:

- ▶ relax the assumption that forecasters know the true data generating process;

To wrap up

New interpretation of forecasts anomalies:

- ▶ relax the assumption that forecasters know the true data generating process;
- ▶ they implement Bayesian learning with unobserved component;

To wrap up

New interpretation of forecasts anomalies:

- ▶ relax the assumption that forecasters know the true data generating process;
- ▶ they implement Bayesian learning with unobserved component;
- ▶ anomalies are persistent $\implies$ learning is slow;

To wrap up

New interpretation of forecasts anomalies:

- ▶ relax the assumption that forecasters know the true data generating process;
- ▶ they implement Bayesian learning with unobserved component;
- ▶ anomalies are persistent $\implies$ learning is slow;
- ▶ unit roots in macroeconomic models are very hard to detect and can cause the slow learning.

Posteriors

Let $\theta = (\rho, \gamma, \sigma^2)$, $X_t = (\mu_{1:t}, x_{1:t})$. Sample B $(\theta^{(b)}, X_t^{(b)})_{b=1}^B$ values as follows:

- ▶ sample $(\theta^{(0)}, X_t^{(0)})$ from the priors;
- ▶ for $b = 1, \dots, B$:
 1. $\rho^{(b+1)} | \gamma^{(b)}, \sigma^{(b)}, X_t^{(b)}, y_{1:t} \sim N(\tilde{\mu}_\rho, \tilde{\sigma}_\rho)$;
 2. $\gamma^{(b+1)} | \rho^{(b+1)}, \sigma^{(b)}, X_t^{(b)}, y_{1:t}$ is drawn through Metropolis-Hastings procedure;
 3. $\sigma^{(b+1)} | \gamma^{(b+1)}, \rho^{(b+1)}, X_t^{(b)}, y_{1:t}$ distribution can be updated by conjugate prior techniques;
 4. $\mu_{1:t}^{(b+1)}, x_{1:t}^{(b+1)} | \theta_t^{(b+1)}, y_{1:t}$ updated through Kalman filter sampling.

▶ Back

Forecasts

For each t

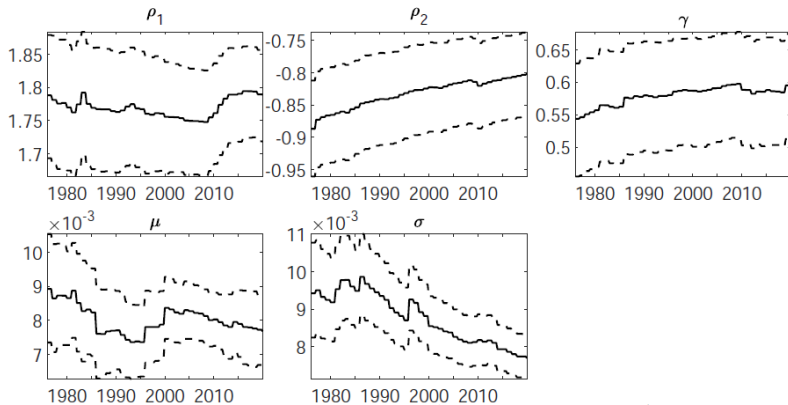
- ▶ for each $b = 1, \dots, B$
 1. generate a path of shocks $(\eta_{t+h}^{(b)}, \omega_{t+h}^{(b)})_{h=1}^H$;
 2. compute $(\mu_{t+h|t}^{(b)}, x_{t+h|h}^{(b)})_{h=1}^H$ using shocks and $\theta_t^{(b)}$;
 3. compute the distribution of the forecasts $(y_{t+h|t}^{(b)})_{h=1}^H$;
- ▶ $\mathbb{F}_t y_{t+h} = \frac{1}{B} \sum_{b=1}^B y_{t+h|t}^{(b)}$

▶ Back

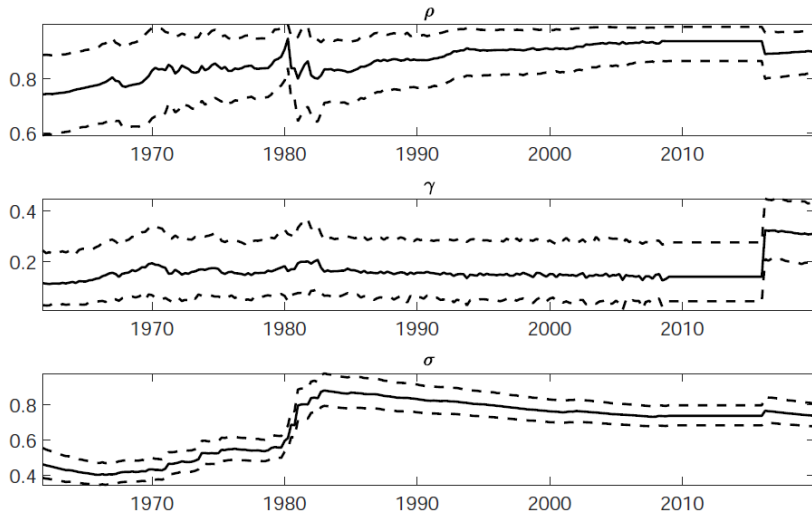
Unobserved Components Model for GDP

$$\begin{aligned}y_t &= z_t + x_t \\ \Delta z_t &= \mu + \sqrt{\gamma}\sigma u_t & u_t &\sim N(0, 1) \\ x_t &= \rho_1 x_{t-1} + \rho_2 x_{t-2} + \sqrt{1 - \gamma}\sigma v_t & v_t &\sim N(0, 1)\end{aligned}$$

		Forecast Horizon				
		1	2	3	4	5
<i><u>Panel A: Bias</u></i>						
	CBO	0.27 (0.21)	-0.27 (0.34)	-0.54 (0.44)	-0.62 (0.47)	-0.52 (0.44)
	UC Model	-0.65** (0.27)	-1.65*** (0.40)	-1.36*** (0.38)	-0.85** (0.38)	-0.66* (0.39)
<i><u>Panel B: Autocorrelation</u></i>						
	CBO	0.22 (0.16)	0.16 (0.17)	0.11 (0.14)	0.08 (0.18)	0.08 (0.11)
	UC Model	0.39** (0.17)	0.31* (0.16)	0.23** (0.11)	0.06 (0.12)	-0.05 (0.07)
<i><u>Panel C: Mincer-Zarnowitz</u></i>						
	CBO	0.94 (0.12)	0.60 (0.31)	0.03*** (0.31)	-0.42*** (0.22)	-0.43*** (0.36)
	UC Model	0.84 (0.14)	0.35*** (0.20)	0.34*** (0.33)	-0.38*** (0.26)	-0.98*** (0.63)
<i><u>Panel D: Coibion-Gorodnichenko</u></i>						
	CBO	0.08 (0.11)	-0.00 (0.23)	0.50 (0.52)	-1.63*** (0.41)	-1.46*** (0.45)
	UC Model	0.06 (0.10)	-0.76 (0.56)	-0.11 (0.29)	-0.78 (0.56)	-1.22** (0.50)

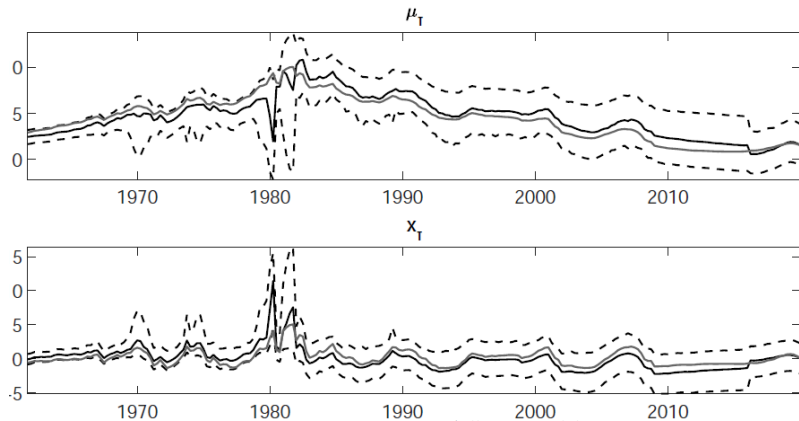


Interest rate Model Performance: parameters dynamics



► Back

Interest rate Model Performance: state dynamics



► Back